of examples. The more diverse and relevant examples you provide, the higher-quality results you will receive (depending on the AI model's performance and maturity).

5.3 Chain-of-thought prompting: A deeper dive into reasoning

In the previous section, we explored the concept of few-shot learning and how it can be used to guide LLMs in generating desired outputs by providing relevant examples. Now we look at another complementary technique called chain-of-thought prompting, which encourages LLMs to break down a problem into smaller, manageable steps and provide a step-by-step explanation of their reasoning process. By explicitly asking the model to show its work, chain-of-thought prompting helps to ensure that the final answer is not only correct but also based on easy-to-verify steps.

As an analogy, imagine a student working on a complex math problem. Instead of simply writing down the final answer, the teacher encourages the student to explain their thought process step by step. In that case, the teacher would reward the student who articulates the answer in a clear way, every step of the way.

5.3.1 Example of chain-of-thought prompting

Let's consider a simple example of using chain-of-thought prompting to solve a math problem that calculates percentages:

```
Prompt: Solve the following problem step-by-step:
What is 25 percent of 80?

Answer:
  1. Understand the question:
     We need to find 25% of 80.
  2. Convert the percentage to a decimal number:
     25% = 25/100 = 0.25
  3. Multiply the decimal by 80:
     0.25 × 80 = 20
Therefore, 25% of 80 is 20 percent.
```

In this example, the prompt's instruction to "Solve the following problem step-by-step" explicitly asks the model to show its work; the model responds by breaking down the problem into three steps and demonstrating its reasoning process.

Of course, just asking the model to break down the steps of each calculation does not guarantee that each step will be correct. There might be cases when the model misinterprets the problem, leading to incorrect intermediate steps or conclusions. This can occur for several reasons:

- *Misunderstanding the problem*—The model may not fully grasp the nuances of the question or the specific requirements of the task.
- *Ambiguous prompts*—Natural language can often be ambiguous, and the way a problem is phrased may lead the model to interpret it differently than intended. Wrong punctuation, minced words, and commas can affect the meaning of the